

Multimodal Hate Video Classification on HateMM

Problem definition

Problem. Hateful content online increasingly travels as video, where meaning is carried jointly by visual context, audio delivery, and the spoken message. Single-modality classifiers struggle on cases where these three signals must be combined to distinguish hate from satire or critical discourse.

Hypothesis. A small transformer trained on top of frozen pretrained per-modality encoders can outperform every unimodal baseline on a real corpus of hateful and non-hateful videos, with gains that are not uniform across target communities.

Dataset

HateMM (Das et al., ICWSM 2023), released under CC BY 4.0 on Zenodo. We use it as-is, without modification or redistribution.

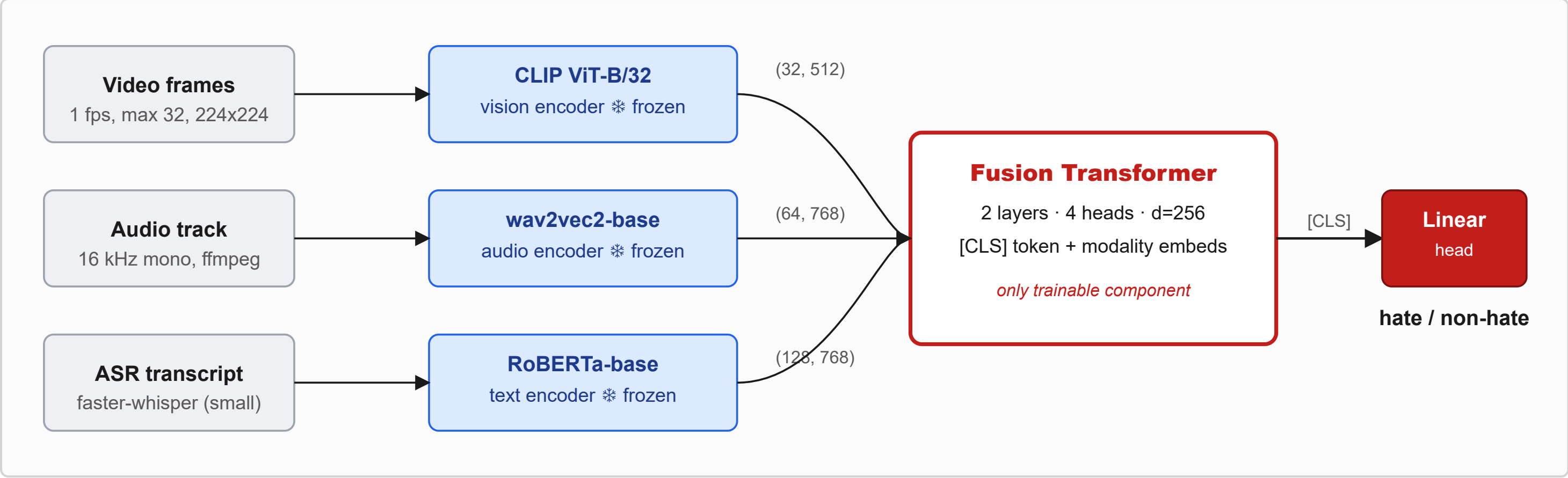
- **1083 BitChute videos** (431 hate, 652 non-hate); ~43.5 hours, 6.3 GB.
- Per-video labels: binary hate flag, target community, hateful-segment timestamps.
- 80/10/10 stratified split (863/109/109), fixed seed; preprocessing succeeded on 1081/1083 videos.

Method

Pipeline in three stages.

1. **Preprocess.** Sample frames at 1 fps capped at 32 per video, center-cropped to 224 x 224; extract 16 kHz mono audio via ffmpeg; transcribe speech with faster-whisper.
2. **Frozen feature extraction.** CLIP ViT-B/32 per frame, wav2vec2-base mean-pooled into 64 audio tokens, RoBERTa-base token embeddings (length 128) on the transcript.
3. **Fusion.** A 2-layer transformer (d = 256, 4 heads, GELU, pre-norm) ingests the concatenated per-modality token sequences plus learnable modality-type embeddings and a [CLS] token; the [CLS] output goes through a linear head for binary classification.

Why frozen encoders. Keeping all three encoders frozen makes the modality ablation a fair comparison: only the small fusion head varies between configurations. Restricting the same architecture to a single modality at the input gives our unimodal baselines.



Related works

- **HateMM [1].** Defines the task; releases the dataset and target-community labels we audit against.
- **CLIP [2].** Provides our frozen vision encoder; we use the image branch only.
- **Multimodal fusion transformers [3].** Motivate the cross-modal-attention design; we adopt a lighter [CLS]-readout variant suited to the dataset scale.

Reproducibility

AdamW (lr 1e-4, batch 16, BCE), early stop on val macro-F1; 3 seeds {42, 1337, 2025} with the split held fixed. Single NVIDIA V100 on EPFL Noto, using the EE-559 course kernel plus *opencv-python-headless* and *faster-whisper*. Full pipeline and aggregator at github.com/YDIB11/Mini-Project.

References

1. M. Das, R. Raj, P. Saha, B. Mathew, M. Gupta, and A. Mukherjee, "HateMM: A Multi-Modal Dataset for Hate Video Classification," in *Proc. AAAI ICWSM*, vol. 17, 2023, pp. 1014-1023.
2. A. Radford et al., "Learning Transferable Visual Models From Natural Language Supervision," in *Proc. ICML*, vol. 139, 2021, pp. 8748-8763.
3. A. Nagrani, S. Yang, A. Arnab, A. Jansen, C. Schmid, and C. Sun, "Attention Bottlenecks for Multimodal Fusion," in *Proc. NeurIPS*, vol. 34, 2021, pp. 14200-14213.

Validation

Table 1. Modality ablation on the test split (n = 109), mean ± std over 3 seeds {42, 1337, 2025}.

Modality	Acc	F1	Prec	Rec	AUROC
V + A + T	0.79 ± 0.05	0.78 ± 0.05	0.74 ± 0.08	0.74 ± 0.08	0.88 ± 0.01
Text only	0.79 ± 0.02	0.78 ± 0.02	0.75 ± 0.06	0.73 ± 0.08	0.83 ± 0.02
Video only	0.72 ± 0.03	0.71 ± 0.02	0.64 ± 0.06	0.68 ± 0.06	0.81 ± 0.01
Audio only	0.67 ± 0.02	0.67 ± 0.03	0.56 ± 0.02	0.71 ± 0.05	0.77 ± 0.03

Table 2. Per-target-community test accuracy (n ≥ 15), mean ± std over 3 seeds.

Community	n	Fusion	Video	Audio	Text
Blacks	46	0.75 ± 0.05	0.70 ± 0.04	0.59 ± 0.06	0.73 ± 0.03
Others	39	0.84 ± 0.01	0.74 ± 0.09	0.68 ± 0.11	0.85 ± 0.05
Jews	15	0.73 ± 0.13	0.67 ± 0.07	0.71 ± 0.04	0.76 ± 0.04

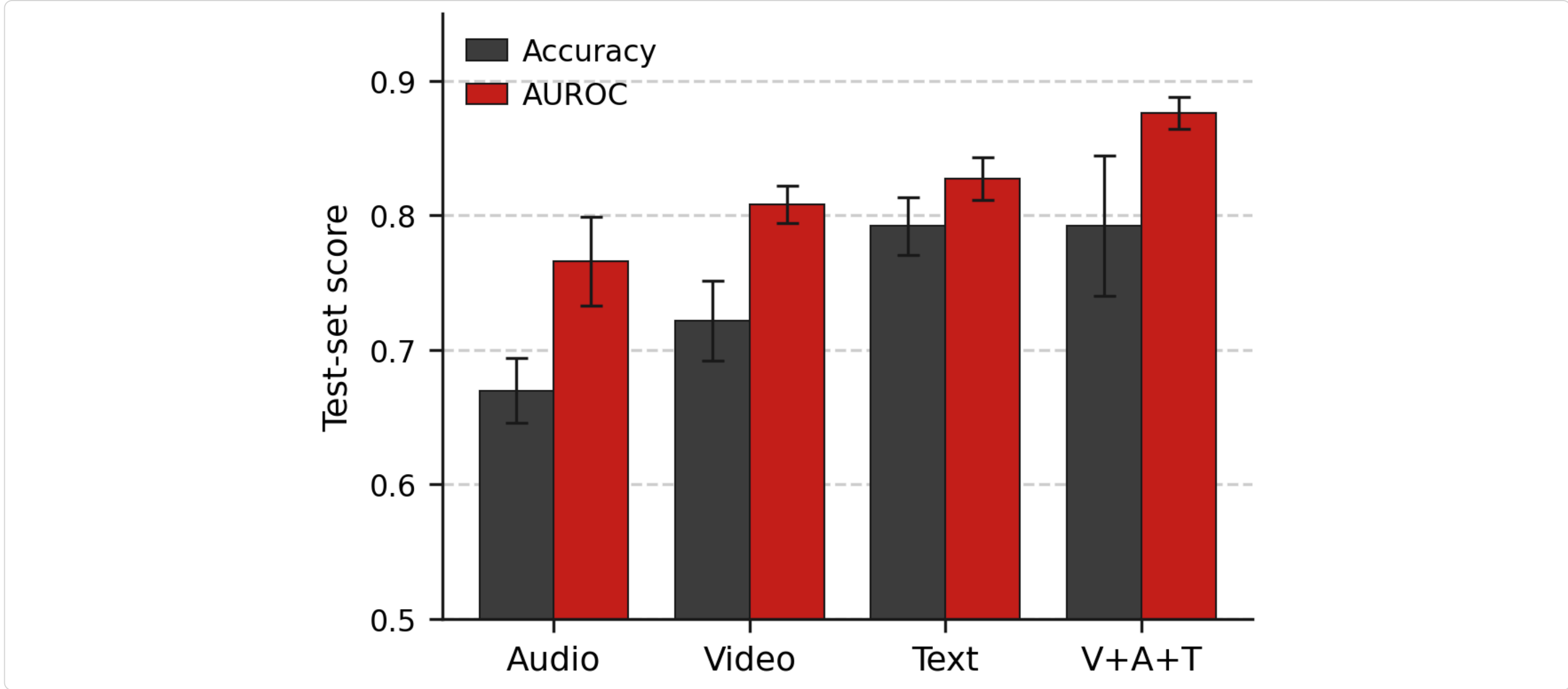


Figure 1. Test accuracy and AUROC across the four models, mean ± std over 3 seeds, sorted ascending by AUROC.

Key observations

Robust ranking gain, noisy threshold gain. Fusion wins AUROC by 5 points over the strongest unimodal (text), with the gap exceeding twice the combined standard deviation. On threshold-dependent metrics (accuracy, F1) fusion matches text on average with markedly higher per-seed variance (std 0.05 vs 0.02).

Text is the strongest unimodal baseline in this dataset, consistent with the verbal nature of much hate content.

Per-community, all fusion-vs-text gaps lie within combined seed variance. On the smallest group (*Jews*, $n = 15$) the fusion model's seed std (0.13) *exceeds* its mean advantage by an order of magnitude.

Limitations

- **Small test set (n = 109).** Per-community stats for groups with $n < 15$ are excluded; reported groups still carry appreciable variance.
- **Heuristic community-label normalisation.** Multi-target labels canonicalised to a sorted comma-joined form, conflating intersectional rows with singleton ones.
- **ASR quality varies.** Shouting, music, and accented speech degrade the text modality on a subset.
- **Architectural choices.** Frozen backbones cap absolute performance, and wav2vec2 hidden states are pooled to a fixed 64 tokens (over-averaging long videos).

Conclusion

Fusing three frozen pretrained encoders through a small transformer head **robustly improves AUROC** over every unimodal baseline on HateMM (0.88 ± 0.01 vs 0.83 ± 0.02 for text). On threshold-dependent metrics the gain washes out into seed variance, a finding single-seed reporting would have missed. Per-community, text alone is competitive with or better than fusion on every reported group, so fusion delivers a calibrated ranking gain rather than a uniform threshold-level improvement over a strong text-only baseline.